

Approximation of Exact Boundary Controllability Problems for the 1-D Wave Equation by Optimization-Based Methods

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ABSTRACT. Optimization-based numerical methods for exact boundary controllability problems for the wave equation are studied. The optimization problems are introduced in order to define unique solutions of the controllability problem. Several properties related to finite difference implementations of the methods are analyzed. Efficient implementation strategies are discussed and the convergence properties of the methods are illustrated through computational experiments. It is shown that for smooth, minimum L^2 -norm Dirichlet controls, the method results in convergent approximations without the need to introduce regularization. Convergent approximations are also obtained for the generic cases of non-smooth minimum L^2 -norm Dirichlet controls and smooth minimum H^1 -norm Dirichlet controls. One of the strengths of the methodology is the flexibility it allows for treating other controls and other minimization criteria; such generalizations are discussed.

1. Introduction

Let T denote a given positive number and let $u_0(\mathbf{x})$ and $u_1(\mathbf{x})$ denote given functions defined on $(0, 1)$. Let $\Sigma = \{0, 1\} \times (0, T)$, $Q = (0, 1) \times (0, T)$ and $(u_0, u_1) \in L^2(0, 1) \times H^{-1}(0, 1)$. The *exact Dirichlet boundary controllability* problem for the wave equation is: *find a control function $g(\mathbf{x}, t)$ defined on Σ and a corresponding state $u(x, t)$ defined on Q such that*

$$(1) \quad \begin{cases} u_{tt} - u_{xx} = 0 & \text{in } Q \\ u|_{t=0} = u_0 \quad \text{and} \quad u_t|_{t=0} = u_1 & \text{in } (0, 1) \\ u|_{t=T} = 0 \quad \text{and} \quad u_t|_{t=T} = 0 & \text{in } (0, 1) \\ u = g & \text{on } \Sigma. \end{cases}$$

It is well known that state and control functions u and g such that (1) is satisfied exist provided T is sufficiently large; see, e.g., [7, 8, 10, 11]. It is worth noting that a computational solution may be obtained for any value of T since finite-dimensional systems are in general exactly controllable [12]; however, for too small

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values of T , the computational solutions will not converge as the temporal and spatial grid sizes tend to zero.

In general, solutions of the controllability problem (1), when they exist, are not unique. However, using the *Hilbert Uniqueness Method* (HUM) introduced in [7, 8], it can be shown that a solution of the controllability problem (1) having minimum $L^2(\Sigma)$ -norm is unique.

More generally, one can consider the *exact boundary controllability problem* (1) with the boundary condition $u = g$ on Σ replaced by $B(u) = g$ on Σ , where B denotes a boundary operator. For example, for $B(u) = (\partial_n u)|_\Sigma = g$, one has the *Neumann* exact boundary controllability problem for the wave equation. One can also consider more general criteria for extracting a particular solution of the controllability problem from the set of all solutions. For example, one can replace the minimum $L^2(\Sigma)$ -norm requirement for the control function g by requiring that the solution minimize a norm-like functional $\mathcal{H}(u)$ over an appropriate function space.

Using a HUM-based approach, a numerical method was developed in [3, 7] to determine approximations of the minimum $L^2(\Sigma)$ -norm solution of the exact Dirichlet boundary controllability problem (1). Numerical implementation were effected through a combination of a finite element method for the spatial discretization and finite difference method for the temporal discretization. However, the computational experiments in [3] showed that the approximate solutions obtained in this manner do not converge to the exact solutions as the temporal and spatial grid sizes tend to zero. Subsequently, a Tychonoff regularization method was added to the solution procedure after which convergent approximations were obtained, at least for smooth controls. The numerical approximation of the controllability problem (1) was also studied in [6, 9, 14]. In these papers, the convergence of the boundary controls for finite difference discretized equations to the control of the continuous 1-D wave equation as the mesh size tends to zero was proved if the step size in time equals to that in space. Furthermore, it was shown in these papers that if the step size in time is less than that in space, that there exist a sequence of controls for the discretized equations that do not converge. Methods of regularization including filtering that result in convergent approximations were also discussed. We will return to these issues later in the paper.

Another computational approach was proposed in [5]. One starts by discretizing the wave equation and the initial and terminal conditions in (1), ignoring the boundary condition. This results in an *underdetermined* linear system of algebraic equations, any of whose solutions provides an approximation to the solution of the controllability problem (1), where the approximate control is simply found by restricting the approximate solution to the boundary. The same approximate solution may be used to determine other controls as well. For example, if a Neumann control is desired, then one merely evaluates $B_h(\vec{U})$ at the boundary, where $B_h(\cdot)$ is an approximation to the normal derivative at the boundary. In addition, there is no difficulty in applying control on only part of the boundary, say at $x = 0$ and applying a specified boundary condition at the remaining parts of the boundary, say at $x = 1$. In this case, during the solution process, one merely ignores the control boundary condition at $x = 0$ but includes the specified boundary conditions at $x = 1$. In [5], the normal equation approach was proposed as a means for determining a particular solution of the underdetermined linear system. Of course, the solution obtained in this manner is an approximation to the minimum $L^2(Q)$ -norm

(not $L^2(\Sigma)$ -norm) solution of (1). Also, note that within the linear system, all time and space levels are coupled so that any naive solution strategy is bound to be inefficient.

2. An optimization-based numerical method

We now introduce a method that incorporates a number of refinements into the method introduced in [5]. A particular solution of the underdetermined linear system is defined by directly solving an optimization problem for which the equations of the linear system act as equality constraints. The functional to be optimized is a discretization of a functional which reflects the desired property one wishes for the particular solution of the underdetermined controllability problem. For example, if one desires a solution of the first three equations in (1) that also minimizes a given norm-like functional $\mathcal{H}(u)$, we will determine an approximate solution by solving the discrete optimization problem consisting of minimizing the functional $\mathcal{H}_h(\vec{U})$ subject to $A\vec{U} = \vec{F}$, where $\mathcal{H}_h(\cdot)$ denotes a discretization of $\mathcal{H}(\cdot)$, $\vec{U}$ is a vector of, e.g., nodal values, of the approximation of u , and $A\vec{U} = \vec{F}$ denotes the underdetermined linear system. We focus on the minimum $L^2(\Sigma)$ -norm and minimum $H^1(\Sigma)$ -norm solutions of the controllability problem and on regularized minimum $L^2(\Sigma)$ -norm solutions. It will be clear that, at least from an algorithmic point of view, our approach is applicable to other norm-like functionals. Also, for the sake of concreteness and simplicity, we consider finite difference discretizations of the first three equations in (1). Our algorithm in higher-dimensional settings is discussed in [4].

The strengths of our approach can be summarized as follows.

- No boundary condition on the control part of the boundary needs to be specified during the solution process; as a result, a single solution can be used to define approximate controls of different types, e.g., Dirichlet or Neumann, by simply evaluating, at the control part of the boundary, an appropriate discrete operator applied to the discrete solution.
- Approximations can be obtained to the solutions of the exact controllability problem that satisfy arbitrary optimization properties, not just the minimum $L^2(\Sigma)$ -norm property.
- When the exact minimum $L^2(\Sigma)$ -norm solution for the controllability problem is smooth, convergence of the approximate controls is achieved, without the need for regularization, for any ratio of time and space step sizes satisfying the CFL condition.
- For the generic case of non-smooth minimum $L^2(\Sigma)$ -norm exact solutions for the controllability problem, convergence can be achieved through a simple regularization of the functional.
- Approximations of minimum $H^1(\Sigma)$ -norm solutions for the controllability problem converge without the need for regularization.

2.1. Coupled space-time finite difference discretization. Let $\Omega = (0, 1)$. Then, the minimum value of T for which exact controllability holds is 1 so that, throughout, we assume that $T \geq 1$. We employ a finite difference discretization for the wave equation. Let $x_i = i/J$, $i = 0, 1, \dots, J$ and $t_k = kT/K$, $k = 0, 1, \dots, K$, where J and K are chosen positive integers, denote a partition of $\Omega \times (0, T)$, and let $u_{i,k}$ denote the approximation to $u(x_i, t_k)$. The grid sizes in space and time are then given by $h = 1/J$ and $\delta = T/K$, respectively. The wave equation (1) is

discretized using standard three-point difference quotients [13] so that the $u_{i,k}$'s are required to satisfy the following discrete wave equation: for $i = 1, 2, \dots, J-1$ and $k = 1, 2, \dots, K-1$,

$$(2) \quad \frac{u_{i,k+1} - 2u_{i,k} + u_{i,k-1}}{\delta^2} - \frac{u_{i+1,k} - 2u_{i,k} + u_{i-1,k}}{h^2} = 0.$$

Let $\lambda = \delta/h$; then (2) can be simplified to: for $i = 1, 2, \dots, J-1$ and $k = 1, 2, \dots, K-1$,

$$(3) \quad u_{i,k+1} - 2(1 - \lambda^2)u_{i,k} - \lambda^2(u_{i+1,k} + u_{i-1,k}) + u_{i,k-1} = 0.$$

The standard CFL condition that guarantees the stability of the scheme (3) is given by $\lambda \leq 1$ which implies that $TJ/K \leq 1$. Then, since $T \geq 1$, we have $K \geq J$. The initial and terminal conditions are discretized by: for $i = 0, 1, \dots, J$,

$$(4) \quad u_{i,0} = u_0(x_i), \quad u_{i,1} = u_{i,0} + \delta u_1(x_i), \quad u_{i,K} = 0, \quad \text{and} \quad u_{i,K-1} = 0.$$

The linear system formed by (3) and (4) has $p = (J-1)(K-1) + 4(J+1)$ equations and $q = (J+1)(K+1)$ unknowns. We symbolically express it in the form $A\vec{U} = \vec{F}$, where the matrix $A \in \mathbb{R}^{p \times q}$ has bandwidth proportional to $J+1$, $\vec{U} \in \mathbb{R}^q$ is a vector of the unknowns $u_{i,k}$, and $\vec{F} \in \mathbb{R}^p$ is a known vector determined from the initial data. It is also worth noting that the number of non-zero elements of each row or each column of A is less than or equal to 5. Furthermore, we have that $q - p = (J+1)(K-3) - (J-1)(K-1) = 2(K-J) - 4$ so that, if we choose $K > J+2$, then $q - p = 2(K-J) - 4 > 4 - 4 > 0$, i.e., the linear system $A\vec{U} = \vec{F}$ has more unknowns than equations.

The unknowns associated with the domains of dependence of the initial and terminal data are uniquely determined. In fact, after computing (4), some of the equations in (3) allow us to uniquely determine those additional unknowns through an explicit marching procedure as follows: without loss of generality, we assume that $J = 2M+1$ for some integer $M > 0$ (the case of J being an even integer can be treated in a similar manner) and implement

$$(5) \quad \begin{aligned} &\text{for } k = 1, \dots, M \\ &\quad k_1 = k, \quad k_2 = K - k \\ &\quad \text{for } i = k, \dots, J - k \\ &\quad \quad u_{i,k_1+1} = 2(1 - \lambda^2)u_{i,k_1} + \lambda^2(u_{i+1,k_1} + u_{i-1,k_1}) - u_{i,k_1-1} \\ &\quad \quad u_{i,k_2-1} = 2(1 - \lambda^2)u_{i,k_2} + \lambda^2(u_{i+1,k_2} + u_{i-1,k_2}) - u_{i,k_2+1} \\ &\quad \text{end} \\ &\text{end} \end{aligned}$$

Let $\vec{V}$ be the vector of the unknowns determined by (4) and (5); it is easily determined that $\vec{V} \in \mathbb{R}^r$, where $r = 4(J+1) + (J^2-1)/2$ is also the number of equations in (4) and (5) together.

Let $\tilde{p} = p - r = (J-1)(K-1) - \frac{1}{2}(J^2-1)$ and $\tilde{q} = q - r = (J+1)(K-3) - \frac{1}{2}(J^2-1)$. Clearly, $q - p = \tilde{q} - \tilde{p}$. Let $\tilde{\vec{U}} \in \mathbb{R}^{\tilde{q}}$ be the vector of $u_{i,k}$'s not belonging to $\vec{V}$. Then, the linear system $A\vec{U} = \vec{F}$ can be reduced to $\tilde{A}\tilde{\vec{U}} = \tilde{\vec{F}}$, where $\tilde{A} \in \mathbb{R}^{\tilde{p} \times \tilde{q}}$ and $\tilde{\vec{F}} \in \mathbb{R}^{\tilde{q}}$ are obtained by deleting the equations used in (4) and (5) to pre-determine $\vec{V}$ and then, in the remaining equations, moving the terms involving the components of $\vec{V}$ to the right-hand side.

The following result implies that the matrix $\tilde{A}$ (and therefore A as well) is of full row rank, i.e., the equations included in the linear system $\tilde{A}\tilde{U} = \tilde{F}$ are linearly independent.

PROPOSITION 1. $\text{rank}(\tilde{A}) = \tilde{p}$.

PROOF. Recall that $J = 2M + 1$. Let $\tilde{W} = \{u_{i,k} \mid i = M, M+1, k = M+2, \dots, K-M-2\}$. One easily finds that $\tilde{W} \in \mathbb{R}^s$, where $s = 2(K-2M-3)$. It is not difficult to verify that $s = \tilde{q} - \tilde{p}$ so that the cardinality of $\tilde{W}$ is exactly the difference between the number of columns and rows of A or $\tilde{A}$. (Since $\tilde{q} > \tilde{p}$, $\tilde{A}\tilde{U} = \tilde{F}$ is an *underdetermined* linear system.)

Assume that $\tilde{W}$ is known. Since $\tilde{V}$ has already been determined and is involved in $\tilde{F}$ and $K > J + 2 > 2(M+2)$ for $M > 0$, we can re-express the linear system $\tilde{A}\tilde{U} = \tilde{F}$ by the following recursive computations, i.e., by a marching procedure from the center of Q to the lateral surface Σ :

$$\begin{aligned}
 & \text{for } i = M, \dots, 1 \\
 & \quad i_1 = i, i_2 = J - i \\
 & \quad \text{for } k = i + 1, \dots, K - i - 1 \\
 & \quad \quad u_{i_1-1,k} = \frac{1}{\lambda^2}(u_{i_1,k+1} + u_{i_1,k-1}) - 2\left(\frac{1}{\lambda^2} - 1\right)u_{i_1,k} - u_{i_1+1,k} \\
 & \quad \quad u_{i_2+1,k} = \frac{1}{\lambda^2}(u_{i_2,k+1} + u_{i_2,k-1}) - 2\left(\frac{1}{\lambda^2} - 1\right)u_{i_2,k} - u_{i_2-1,k} \\
 & \quad \text{end} \\
 & \text{end}
 \end{aligned} \tag{6}$$

By the above procedure, we finally have determined all the components of $\tilde{U}$ in the linear system $\tilde{A}\tilde{U} = \tilde{F}$ and we have thus shown that $\text{rank}(\tilde{A}) = \tilde{q} - s = \tilde{p}$, i.e., by setting s of the components in $\tilde{U}$, $\tilde{A}\tilde{U} = \tilde{F}$ uniquely determines the remaining components. Thus, $\tilde{A}$ has full row rank. $\square$

2.2. Constrained quadratic optimization problems. We now turn to the selection of a particular solution of the linear system $A\tilde{U} = \tilde{F}$, i.e., of the discretized wave equation and initial and terminal conditions, by requiring that solution to also solve a discrete optimization problem.

2.2.1. *Minimum $L^2(\Sigma)$ -norm solutions.* We first focus on finding approximations of the minimum $L^2(\Sigma)$ -norm solution of the controllability problem. Specifically, we use the trapezoidal rule to obtain the approximation

$$\begin{aligned}
 \mathcal{J}(u) & \equiv \|g\|_{L^2(\Sigma)}^2 = \int_{\Sigma} u^2 d\Gamma dt = \int_0^T [u^2(0,t) + u^2(1,t)] dt \\
 & \approx \delta \sum_{k=0}^K \alpha_k (u_{0,k}^2 + u_{J,k}^2) \equiv \mathcal{J}_h(\tilde{U}),
 \end{aligned} \tag{7}$$

where $\alpha_0 = \alpha_K = 1/2$ and $\alpha_1 = \dots = \alpha_{K-1} = 1$. Furthermore, we effect the partition $\tilde{U} = (\tilde{U}_1^T, \tilde{U}_2^T)^T$, where $\tilde{U}_1$ denotes the subvector of $\tilde{U}$ belonging to the boundary $\bar{\Sigma}$ and $\tilde{U}_2$ denotes the subvector of $\tilde{U}$ belonging to interior Q . It is seen

that $\tilde{\tilde{U}}_1 \in \mathbb{R}^{\tilde{q}_1}$, where $\tilde{q}_1 = 2(K-3)$, and $\tilde{\tilde{U}}_2 \in \mathbb{R}^{\tilde{q}_2}$, where $\tilde{q}_2 = \tilde{q} - 2(K-3)$. We are then led to the following *discrete* quadratic optimization problem:

$$(8) \quad \min_{\tilde{\tilde{U}} \in \mathbb{R}^{\tilde{q}}} \mathcal{J}_h(\tilde{\tilde{U}}) = \delta(\tilde{\tilde{U}}_1^T \tilde{\tilde{U}}_1 + b) \quad \text{subject to} \quad \tilde{A}_1 \tilde{\tilde{U}}_1 + \tilde{A}_2 \tilde{\tilde{U}}_2 = \tilde{\tilde{F}},$$

where $\tilde{A}_1 \in \mathbb{R}^{\tilde{p} \times \tilde{q}_1}$ and $\tilde{A}_2 \in \mathbb{R}^{\tilde{p} \times \tilde{q}_2}$ are obtained by appropriately rearranging and partitioning $\tilde{A}$ corresponding to $\tilde{\tilde{U}}_1$ and $\tilde{\tilde{U}}_2$, and b is a constant depending on the components of $\tilde{U}$ belonging to $\bar{\Sigma}$ that are pre-determined by (4). It should be pointed out that the problem (8) couples space and time together. Regarding the solution of the optimization system (8), we have the following result.

PROPOSITION 2. *The discrete quadratic optimization problem (8) has a unique solution.*

PROOF. Existence is trivial since $\mathcal{J}(\tilde{\tilde{U}})$ is a convex functional of $\tilde{\tilde{U}}$ and, by Proposition 1, $\tilde{A}\tilde{\tilde{U}} = \tilde{\tilde{F}}$ is a set of linearly independent constraints. Now, suppose that both $\tilde{\tilde{U}} = (\tilde{\tilde{U}}_1, \tilde{\tilde{U}}_2)$ and $\tilde{\tilde{Z}} = (\tilde{\tilde{Z}}_1, \tilde{\tilde{Z}}_2)$ are solutions of the optimization problem (8). Then, $\mathcal{J}_h(\tilde{\tilde{U}}) = \delta(\tilde{\tilde{U}}_1^T \tilde{\tilde{U}}_1 + b) = \delta(\tilde{\tilde{Z}}_1^T \tilde{\tilde{Z}}_1 + b) = \mathcal{J}_h(\tilde{\tilde{Z}})$. If we define $\mathcal{J}_1(\tilde{\tilde{U}}_1) = \delta(\tilde{\tilde{U}}_1^T \tilde{\tilde{U}}_1 + b)$ for $\tilde{\tilde{U}}_1 \in \mathbb{R}^{\tilde{q}_1}$, it is easy to show that $\mathcal{J}_1$ is a strictly convex function of $\tilde{\tilde{U}}_1$. Then, the unconstrained minimization problem of minimizing $\mathcal{J}_1(\tilde{\tilde{U}}_1)$ over $\tilde{\tilde{U}}_1 \in \mathbb{R}^{\tilde{q}_1}$ has a unique solution and we conclude that $\tilde{\tilde{U}}_1 = \tilde{\tilde{Z}}_1$. With the same discrete initial conditions (4), we then obtain that $\tilde{\tilde{U}}_2 = \tilde{\tilde{Z}}_2$ by explicit marching. Thus, $\tilde{\tilde{U}} = \tilde{\tilde{Z}}$. $\square$

The HUM control of the exact Dirichlet boundary controllability problem (1) is equivalent to the following continuous optimization problem [7]:

$$(9) \quad \min_{u|_{\Sigma} \in L^2(\Sigma)} \mathcal{J}(u) = \|g\|_{L^2(\Sigma)}^2 \quad \text{subject to} \quad (1).$$

We then have the following result.

PROPOSITION 3. *Let u be the solution of problem (9) and u^h the corresponding discrete solution of (7) with $\delta = h$, i.e., $\lambda = 1$. Then, $g^h \rightarrow g$ strongly in $L^2(\Sigma)$, where $g = u|_{\Sigma}$ and $g^h = u^h|_{\Sigma}$.*

See [9] for the proof. In fact, the convergence of the discrete Dirichlet control for this special mesh ration result from the stability condition $\lambda = \frac{1}{\lambda} = 1$ and the symmetry of 1-D wave equation, i.e., marching is stable for both the time and space directions under such a mesh restriction.

2.2.2. Minimum $H^1(\Sigma)$ -norm and regularized minimum $L^2(\Sigma)$ -norm solutions. We also consider finding the solution of the exact controllability problem (1) that has minimum $H^1(\Sigma)$ -norm; controls g determined in this manner are more regular than those obtained through the minimization of the $L^2(\Sigma)$ -norm. For the minimum

$H^1(\Sigma)$ -norm case, we use the approximation

$$(10) \quad \begin{aligned} \mathcal{K}(u) &\equiv \|g\|_{H^1(\Sigma)}^2 = \int_0^T [u^2|_{(0,t)} + u_t^2|_{(0,t)} + u^2|_{(1,t)} + u_t^2|_{(1,t)}] dt \approx \mathcal{K}_h(\vec{U}) \\ &\equiv \delta \sum_{k=0}^K \alpha_k (u_{0,k}^2 + u_{J,k}^2) + \frac{1}{\delta} \sum_{k=1}^K \left((u_{0,k} - u_{0,k-1})^2 + (u_{J,k} - u_{J,k-1})^2 \right), \end{aligned}$$

and are led to another discrete quadratic optimization problem:

$$(11) \quad \min_{\vec{U} \in \mathbb{R}^{\tilde{q}}} \mathcal{K}_h(\vec{U}) = \delta(\vec{U}_1^T R \vec{U}_1 + b) \quad \text{subject to} \quad \tilde{A}_1 \vec{U}_1 + \tilde{A}_2 \vec{U}_2 = \vec{F},$$

where $R \in \mathbb{R}^{\tilde{q}_1 \times \tilde{q}_1}$ is a tridiagonal, positive definite matrix. The result analogous to Proposition 2 still holds in this case; in fact, one may view (8) as the special case of (11) for which R is the identity matrix.

If there exists a solution $u|_{\Sigma} \in H^1(\Sigma)$ satisfying (1) and if $\lambda = 1$, then it is easy to see that the convergence of the discrete control still holds, i.e., $g^h \rightarrow g$ in $L^2(\Sigma)$, where g is the control of problem (1) having minimum $H^1(\Sigma)$ -norm and g^h is the corresponding discrete control.

More generally, we could, for $\epsilon > 0$, define and approximate the “hybrid” functional

$$(12) \quad \begin{aligned} \mathcal{K}_\epsilon(u) &\equiv \|g\|_{L^2(\Sigma)}^2 + \epsilon \|g\|_{H^1(\Sigma)}^2 \approx \mathcal{K}_{\epsilon,h}(\vec{U}) \\ &\equiv \delta \sum_{k=0}^K \alpha_k (u_{0,k}^2 + u_{J,k}^2) + \frac{\epsilon}{\delta} \sum_{k=1}^K \left((u_{0,k} - u_{0,k-1})^2 + (u_{J,k} - u_{J,k-1})^2 \right), \end{aligned}$$

where $|\cdot|_{H^1(\Sigma)}^2$ denotes the seminorm, and then find the control g that minimizes the norm-like functional (12). It is clear that (7) and (10) are only special cases of (12), i.e., $\mathcal{K}_0(\cdot) = \mathcal{J}(\cdot)$ and $\mathcal{K}_1(\cdot) = \mathcal{K}(\cdot)$. For $0 < \epsilon \ll 1$, one can view $\mathcal{K}_\epsilon(\cdot)$ as a regularization of $\mathcal{J}(\cdot)$.

2.3. Solution processes and implementation.

A Lagrange multiplier method. If $\vec{U}$ is the solution of the optimization problems (8) or (11), then the Lagrange multiplier rule states that $\vec{U}$ is a stationary point of the Lagrangian functional $\mathcal{L}(\vec{U}) = \delta(\vec{U}_1^T R \vec{U}_1 + b) - \vec{S}^T (\tilde{A}_1 \vec{U}_1 + \tilde{A}_2 \vec{U}_2 - \vec{F})$, where $\vec{S} \in \mathbb{R}^{\tilde{p}}$ is the Lagrange multiplier introduced to enforce the constraint equations $\tilde{A}_1 \vec{U}_1 + \tilde{A}_2 \vec{U}_2 = \vec{F}$ and $R = I$ in case of (8). Then, it is easily shown that $\vec{U} = (\vec{U}_1, \vec{U}_2)$ and $\vec{S} = \vec{S}/\delta$ are the solution of

$$(13) \quad \begin{pmatrix} 0 & \tilde{A}_2^T \\ \tilde{A}_2 & -\tilde{A}_1^T R^{-1} \tilde{A}_1 \end{pmatrix} \begin{pmatrix} \vec{U}_2 \\ \vec{S} \end{pmatrix} = \begin{pmatrix} 0 \\ \vec{F} \end{pmatrix} \quad \text{and} \quad \vec{U}_1 = -R^{-1} \tilde{A}_1^T \vec{S}.$$

Note that $\vec{U}_1$ along with the boundary $u_{i,k}$'s found from (4) and (5) can be used to define the approximation to the exact Dirichlet boundary control $u|_{\Sigma} = g$. Clearly, the coefficient matrix of the linear system in the left equation in (13) is symmetric but it is not positive definite. Note that $\tilde{A}_1$ and $\tilde{A}_2$ are sparse matrices having, like A , at most 5 non-zero elements in each of their rows or columns. Thus, it is efficient to use iterative methods, e.g., BiCG, SYMMQL, or CGNR, to solve the first equation in (13). In higher dimensions, the application of a preconditioner

may be useful since the dimension of the the first equation in (13) may be large. In a practical implementation, the matrices $\tilde{A}_1$ and $\tilde{A}_2$ need not be stored. It is also worth noting that, at each step of an iteration, we may only need to update $\tilde{S}$ and that part of $\tilde{W}$ included in $\tilde{U}_2$; then the remainder of $\tilde{U}_2$ can be directly computed by a marching procedure.

Reduction to an unconstrained optimization problem. Another method for solving the optimization problems (8) or (11) uses the marching procedure (6) to determine $\tilde{U}_1$ from $\tilde{W}$, provided that $\tilde{V}$ is pre-calculated using (4) and (5). Thus, $\tilde{U}_1$ is a linear function of $\tilde{W}$, i.e., there exists a matrix $D \in \mathbb{R}^{\tilde{q}_1 \times s}$ such that $\tilde{U}_1 = D\tilde{W} + \tilde{C}$, where $\tilde{C} \in \mathbb{R}^{\tilde{q}_1}$ is determined from $\tilde{V}$. Then, the optimization problem (11) (or (8) if we set $R = I$) is equivalent to $\min_{\tilde{W} \in \mathbb{R}^r} \mathcal{K}_h(\tilde{U}_1)$ subject to $\tilde{U}_1 = D\tilde{W} + \tilde{C}$. Substituting $\tilde{U}_1 = D\tilde{W} + \tilde{C}$ into $\mathcal{K}_h(\tilde{U}_1)$, we obtain $\mathcal{K}_h(\tilde{U}_1) = \tilde{\mathcal{K}}_h(\tilde{W}) = \tilde{W}^T(D^T R D)\tilde{W} + 2\tilde{C}^T R D\tilde{W} + \tilde{C}^T R \tilde{C}$ so that the constrained optimization problem for $\mathcal{K}_h(\tilde{U}_1)$ is equivalent to the *unconstrained* optimization problem $\min_{\tilde{W} \in \mathbb{R}^r} \tilde{\mathcal{K}}_h(\tilde{W})$. Thus, if $\tilde{W}$ is a stationary point of $\tilde{\mathcal{K}}_h(\cdot)$, we obtain $D^T R D\tilde{W} = -2D^T R \tilde{C}$. Since $D^T R D$ is a positive definite matrix, we can use, e.g., the conjugate gradient method, to solve this linear system.

The matrix $D \in \mathbb{R}^{\tilde{q}_1 \times s}$ is a full matrix. In higher dimensions, $\tilde{q}_1$ is a large number even for small J and K so that directly storing D requires large amounts of memory. However, recall that D embodies the marching procedure (6); that procedure is done level by level from the center of the space-time rectangle Q to its lateral sides Σ . In fact, the marching procedure can be expressed as

$$\begin{cases} \tilde{U}_{M-1} = D_{3,M-1}\tilde{W} + \tilde{C}_{M-1} \\ \tilde{U}_{M-2} = D_{2,M-2}\tilde{U}_{M-1} + D_{3,M-2}\tilde{W} + \tilde{C}_{M-2} \\ \tilde{U}_m = D_{1,m}\tilde{U}_{m+1} + D_{2,m}\tilde{U}_{m+2} + \tilde{C}_m \quad \text{for } m = M-3, \dots, 0, \end{cases}$$

where $\tilde{U}_m = \{u_{i,k} \in \tilde{U} \mid i = m \text{ or } i = J - m\}$ for $m = 0, 1, \dots, M-1$ and the $\{D_{i,m}\}$ are matrices. Clearly, $\tilde{U}_1 = \tilde{U}_0$ and each $D_{i,m}$ is a banded matrix having at most 5 non-zero elements in each of its rows or columns and D can be represented by the sequence of matrices $\{D_{i,m}\}$ corresponding to each level of the marching procedure (6). Consequently, we need only save $\{D_{i,m}\}$ to completely determine the action of D and D^T on a vector.

Note that this implementation cannot be directly applied to implicit (in time) approximations of the wave equation or to finite element approximations involving (unlumped) mass matrices. However, its extension to higher dimensions is straightforward.

3. Computational experiments

3.1. An example with known smooth exact solution. A smooth exact solution to the minimum $L^2(\Sigma)$ -norm controllability problem can be constructed by using Fourier series in a similar way to that used in [3]. Suppose that $Q = (0, 1) \times (0, 7/4)$ and $\Sigma = \{0, 1\} \times (0, 7/4)$. Let $\psi_0(x, t) = -\sqrt{2}\pi \cos \pi(t - \frac{1}{4}) \cos 2\pi x$

and

$$\begin{aligned} \psi_1(x, t) = & \left[2\sqrt{2}\pi(T-t)\sin\left(t - \frac{1}{4}\right) - \frac{10}{3}\sqrt{2}\sin\pi(t-T) \right] \sin\pi x \\ & + \sum_{p \geq 3 \text{ and } p \text{ odd}} \frac{4\sqrt{2}}{p^2-1} \left[\frac{3p}{p^2-4}\cos\pi\left(t - \frac{1}{4}\right) + \sin p\pi(t-T) \right] \sin p\pi x. \end{aligned}$$

Then, set the initial conditions

$$(14) \quad u_0(x) = \psi_0(x, 0) + \psi_1(x, 0) \quad \text{and} \quad u_1(x) = \frac{\partial\psi_0}{\partial t}(x, 0) + \frac{\partial\psi_1}{\partial t}(x, 0).$$

The computation of u_0 and u_1 involve the summation of infinite trigonometric series. Figure 1 provides plots of u_0 and u_1 . It is worth noting that u_0 is a Lipschitz continuous function but does not belong to $C^1(\overline{\Omega})$ and u_1 is a bounded function but does not belong to $C^0(\overline{\Omega})$. For the initial data (14), it can be shown that $u(x, t) = \psi_0(x, t) + \psi_1(x, t)$ is the exact solution having minimum $L^2(\Sigma)$ -norm of the controllability problem given by the first three equations in (1). The corresponding exact Dirichlet control g is then given by restricting $u(x, t)$ to the lateral sides Σ , i.e.,

$$(15) \quad g(t) = u(0, t) = u(1, t) = -\sqrt{2}\pi \cos\pi\left(t - \frac{1}{4}\right).$$

For future reference, note that $\|g\|_{L^2(\Sigma)} \approx 6.1340$ and $\|u\|_{L^2(Q)} \approx 5.8656$.

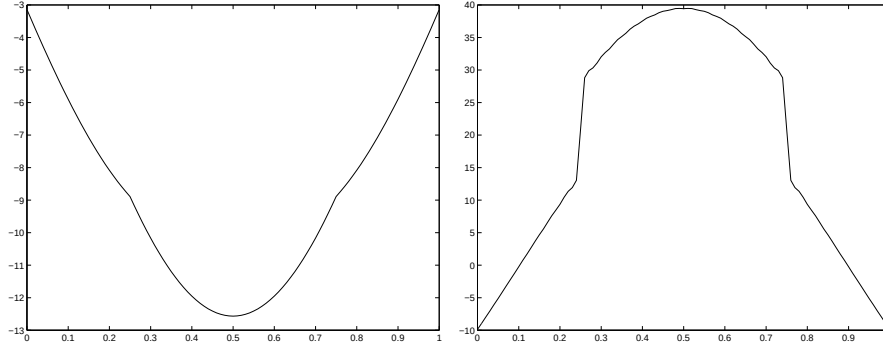


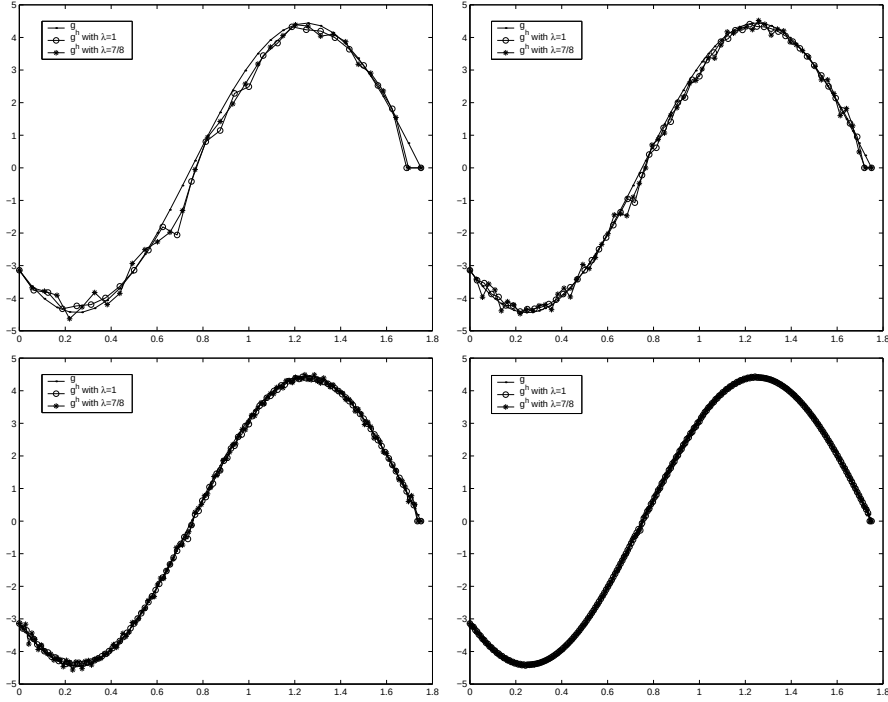
FIGURE 1. The initial data $u_0(\mathbf{x})$ (left) and $u_1(\mathbf{x})$ (right) given in (14).

We apply our numerical method to this example. Computational experiments were carried out for $h = 1/16, 1/32, 1/64$, and $1/128$ with $\lambda = 1$ and $\lambda = 7/8$ respectively, so that the stability condition $K \geq J$ is satisfied.

The results of our computational experiments are summarized in Table 1, where u^h and g^h are the computed approximations of the exact solutions u and g , respectively. All norms were calculated by linearly interpolating the nodal values of u^h and g^h . From this table, it seems that g^h converges to g in the $L^2(\Sigma)$ -norm at a rate of roughly 1 as does u^h to u in the $L^2(Q)$ -norm. In order to visualize the convergence of our method as h becomes smaller, in Figure 2 we provide plots of the exact solution g and the corresponding computed discrete solutions g^h for $h = 1/16, 1/32, 1/64$, and $1/128$.

TABLE 1. Results of computational experiments for the minimum $L^2(\Sigma)$ -norm case for the examples with initial data (14).

h		1/16	1/32	1/64	1/128
$\ g^h\ _{L^2(\Sigma)}$	$\lambda = 1$	5.9572	6.0313	6.0804	6.1008
	$\lambda = 7/8$	5.9803	6.0507	6.0890	6.1104
$\ g^h - g\ _{L^2(\Sigma)} / \ g\ _{L^2(\Sigma)}$	$\lambda = 1$	10.858%	4.625%	2.071%	1.051%
	$\lambda = 7/8$	9.662%	5.003%	2.137%	0.991%
$\ u^h\ _{L^2(Q)}$	$\lambda = 1$	5.6954	5.7873	5.8266	5.8442
	$\lambda = 7/8$	5.7249	5.7899	5.8327	5.8470
$\ u^h - u\ _{L^2(Q)} / \ u\ _{L^2(Q)}$	$\lambda = 1$	9.283%	4.153%	1.989%	1.012%
	$\lambda = 7/8$	7.636%	3.907%	1.864%	0.891%

FIGURE 2. The exact minimum $L^2(\Sigma)$ -norm Dirichlet control g and the approximate controls g^h vs. time along $x = 0$ (and $x = 1$) for the example with initial data (14). Top-left: $h = 1/16$; top-right: $h = 1/32$; bottom-left: $h = 1/64$; bottom-right: $h = 1/128$.

It seems that our method produces (pointwise) convergent approximations for both $\lambda = 1$ and $\lambda = 7/8$ *without the need for regularization*. This should be contrasted with other methods, e.g., that of [3], for which when $\lambda < 1$, regularization was needed in order to obtain convergence. Given Proposition 3, the results for $\lambda = 1$ are not surprising. It may at first seem that our results for $\lambda < 1$ are contradictory to the result of [9] which states that there exists a sequence of approximate

controls that do not converge. However, that result says nothing about the *particular* sequence of minimum discrete $L^2(\Sigma)$ -norm controls that is determined by our method. Thus, the seeming convergence of our sequence of discrete controls is not contradictory.

3.2. Generic examples with minimum $L^2(\Sigma)$ -norm boundary control.

In the example discussed in Section 3.1, the minimum $L^2(\Sigma)$ -norm control is very smooth. Using our methods, we obtained good approximations for this example without the need for regularization. However, this is not the generic case. In general, even for smooth initial data, the minimum $L^2(\Sigma)$ -norm Dirichlet control for the controllability problem (1) will not be smooth. In this section, we illustrate this point and also examine the performance of our method for the generic case.

We again choose $Q = (0, 1) \times (0, 7/4)$ and consider two sets of $C^\infty(\bar{\Omega})$ initial data:

$$(16) \quad \begin{array}{ll} \text{I.} & u_0(x) = \sin(\pi x) \quad \text{and} \quad u_1(x) = \pi \sin(\pi x) \\ \text{II.} & u_0(\mathbf{x}) = e^x \quad \text{and} \quad u_1(\mathbf{x}) = xe^x. \end{array}$$

Note that the initial conditions (I) vanish at the boundary and, that due to symmetry, we have that $u(0, t) = u(1, t)$, i.e., the control at the two sides of Q are the same. For the initial conditions (II), we have that $u(0, t) \neq u(1, t)$.

We examine the case $\lambda = 1$. In Figure 3, we show the results for the control for several grid sizes ranging from $h = 1/16$ to $h = 1/256$. The (pointwise) convergence of the approximations is evident. Note that for the initial conditions given in (16), the minimum $L^2(\Sigma)$ -norm controls are seemingly piecewise smooth, i.e., they contain jump discontinuities. The pointwise convergence of the approximate control for the case of $\lambda = 1$ is probably a one-dimensional artifact; it is likely due to the fact that both the space and time variables in the wave equation in one dimension can act as time-like variables.

Since the exact solutions corresponding to the initial conditions given in (16) are not known, we will take the seemingly converged solutions obtained with $h = 1/256$ and $\lambda = 1$ as a baseline to which we compare our other computational results. We will denote this baseline approximate control by g^* .

Further details about the computational results for the examples with initial conditions given in (16) with $\lambda = 1$ are given in Table 2. The convergence in $L^2(\Sigma)$ of the approximate minimum $L^2(\Sigma)$ -norm controls g^h is evident as is the convergence in $L^2(Q)$ of the approximate solution u^h ; the rates of convergence are seemingly first order.

TABLE 2. Results of computational experiments for the minimum $L^2(\Sigma)$ -norm case for Examples I and II with initial data (16) and for $\lambda = 1$.

	h	1/16	1/32	1/64	1/128	1/256
I	$\ g^h\ _{L^2(\Sigma)}$	0.6831	0.6389	0.6164	0.6048	0.6011
	$\ g^h - g^*\ _{L^2(\Sigma)} / \ g^*\ _{L^2(\Sigma)}$	19.584%	12.591%	7.101%	4.151%	-
	$\ u^h\ _{L^2(Q)}$	0.7642	0.7356	0.7219	0.7151	0.7114
II	$\ g^h\ _{L^2(\Sigma)}$	1.5181	1.3038	1.1876	1.1265	1.0977
	$\ g^h - g^*\ _{L^2(\Sigma)} / \ g^*\ _{L^2(\Sigma)}$	21.288%	13.615%	7.825%	4.237%	-
	$\ u^h\ _{L^2(Q)}$	1.3608	1.2708	1.2278	1.2054	1.1923

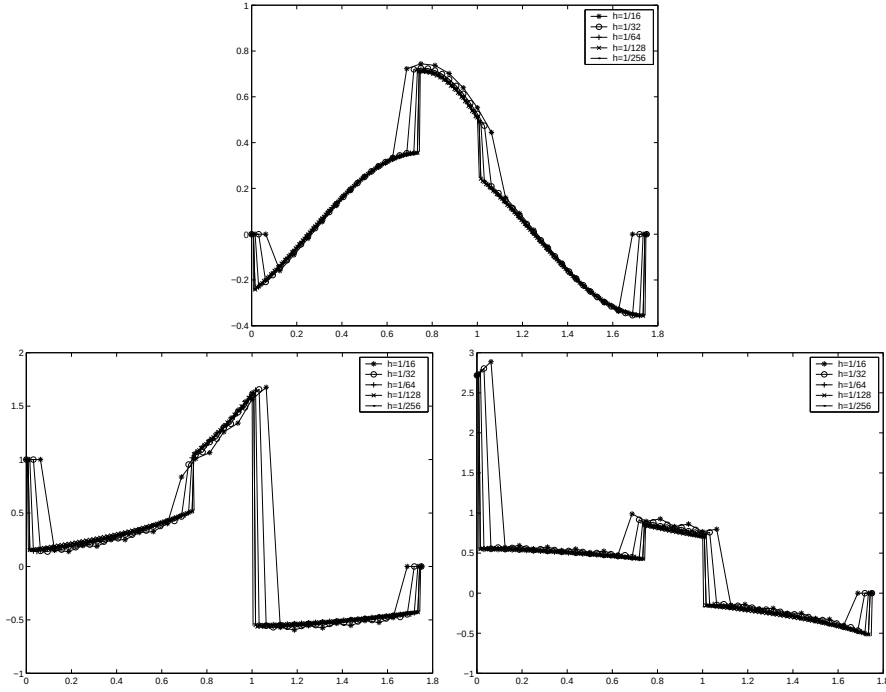


FIGURE 3. The approximate minimum $L^2(\Sigma)$ -norm Dirichlet control g^h vs. time for $\lambda = 1$ for the examples with initial data (16-I) at $x = 0$ and $x = 1$ (top) and (16-II) at $x = 0$ (bottom-left) and $x = 1$ (bottom-right).

Computational experiments were also carried out for $\lambda = 7/8$ and $\lambda = 7/12$ for several values of the grid size ranging from $h = 1/16$ to $h = 1/128$. The results are summarized in Table 3. In Figures 4 and 5, we respectively provide, for the two sets of initial conditions, plots of the computed discrete solution g^h for the two different values of λ and for different values of the grid size.

TABLE 3. Results of computational experiments for the minimum $L^2(\Sigma)$ -norm case for Examples I and II with initial data (16) and for $\lambda < 1$.

		λ	$h = 1/16$	$h = 1/32$	$h = 1/64$	$h = 1/128$
I	$\ g^h\ _{L^2(\Sigma)}$	7/8	0.6715	0.6354	0.6147	0.6036
		7/12	0.6550	0.6212	0.6085	0.5977
	$\ u^h\ _{L^2(Q)}$	7/8	0.7546	0.7327	0.7204	0.7140
		7/12	0.7431	0.7234	0.7164	0.7124
II	$\ g^h\ _{L^2(\Sigma)}$	7/8	1.4584	1.2868	1.1799	1.0933
		7/12	1.3723	1.2157	1.1312	1.1134
	$\ u^h\ _{L^2(Q)}$	7/8	1.3252	1.2569	1.2211	1.1964
		7/12	1.2718	1.2297	1.2077	1.2029

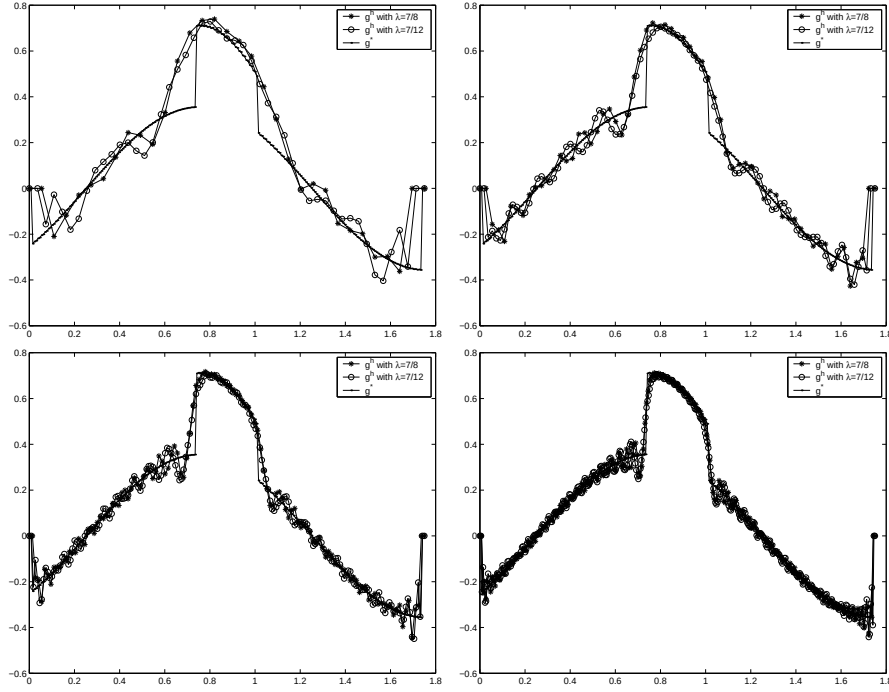


FIGURE 4. The approximate minimum $L^2(\Sigma)$ -norm Dirichlet controls g^h vs. time at $x = 0$ (and $x = 1$) for the initial data (16-I). Top-left: $h = 1/16$; top-right: $h = 1/32$; bottom-left: $h = 1/64$; bottom-right: $h = 1/128$.

From Figures 4 and 5, we see that the approximate minimum $L^2(\Sigma)$ -norm Dirichlet controls obtained with values of $\lambda < 1$ are highly oscillatory. In fact, the frequencies of the oscillations increase with decreasing grid size. However, it seems that the amplitudes of the oscillations do not increase as the grid size decreases. Furthermore, from the results in Table 3, it seems that for $\lambda < 1$, the approximate controls g^h , although oscillatory in nature and nonconvergent in a pointwise sense, converge in an $L^2(\Sigma)$ sense. Also, the solution u^h is seemingly convergent in an $L^2(Q)$ sense.

The results of Table 3 and Figures 4 and 5 indicate that for the generic case of non-smooth minimum $L^2(\Sigma)$ controls and for general $\lambda < 1$, our method produces convergent (in $L^2(Q)$ and $L^2(\Sigma)$) approximations *without the need of regularization* but the approximations are not in general convergent in a pointwise sense. Of course, approximations that do not converge in a pointwise sense may be of little practical use, even if they converge in a root mean square sense.

3.3. Generic examples with minimum $H^1(\Sigma)$ -norm boundary control.

The computational experiments of Section 3.2 indicate that, generically, minimum $L^2(\Sigma)$ -norm boundary controls are only piecewise smooth and that approximations obtained by our method are in general not pointwise convergent so that they are not realizable in practice. In order to obtain smoother controls, i.e., at least continuous in time, we consider the particular solution of the exact controllability problem (1)

having minimum $H^1(\Sigma)$ -norm. We again use the numerical methods discussed in Section 2 and the examples with initial condition given in (16).

Table 4 and Figures 6 and 7 provide the results of computational experiments for minimum $H^1(\Sigma)$ -norm controls. In Table 4, $\tilde{g}$ refers to the approximate minimum $H^1(\Sigma)$ -norm Dirichlet control obtained for $\lambda = 1$ and $h = 1/128$. Figures 6 and 7 indicate that, in general, minimum $H^1(\Sigma)$ -norm solutions of the controllability problem are smooth and that the approximate controls obtained by our method converge in a pointwise sense for both $\lambda = 1$ and $\lambda < 1$ without the need for regularization. The results of Table 4 indicate that our method produces convergent (in $L^2(Q)$ and $L^2(\Sigma)$) approximations.

TABLE 4. Results of computational experiments for the minimum $H^1(\Sigma)$ -norm case for the examples with initial data (16).

		λ	$h = 1/16$	$h = 1/32$	$h = 1/64$	$h = 1/128$
I	$\ g^h\ _{L^2(\Sigma)}$	1	0.8341	0.8040	0.7899	0.7827
		7/8	0.8507	0.7958	0.7868	0.7824
	$\ g^h - \tilde{g}\ _{L^2(\Sigma)} / \ \tilde{g}\ _{L^2(\Sigma)}$	1	7.865%	3.005%	1.101%	–
		7/8	10.885%	2.582%	0.806%	–
	$\ u^h\ _{L^2(Q)}$	1	0.8735	0.8450	0.8311	0.8240
		7/8	0.8818	0.8384	0.8266	0.8230
II	$\ g^h\ _{L^2(\Sigma)}$	1	2.2990	2.2453	2.2300	2.2273
		7/8	2.2650	2.2438	2.2319	2.2283
	$\ g^h - \tilde{g}\ _{L^2(\Sigma)} / \ \tilde{g}\ _{L^2(\Sigma)}$	1	4.999%	1.400%	0.572%	–
		7/8	4.681%	1.331%	0.477%	–
	$\ u^h\ _{L^2(Q)}$	1	1.7471	1.7182	1.7102	1.7077
		7/8	1.7303	1.7179	1.7128	1.7084

3.4. Generic examples with regularized minimum $L^2(\Sigma)$ -norm controls. In Section 3.2, we considered approximations of some generic examples for minimum $L^2(\Sigma)$ -norm controls and found that, in general, approximations converged in an $L^2(\Sigma)$ sense but not pointwise, unless $\lambda = 1$. In Section 3.3, we considered approximations of some generic examples for minimum $H^1(\Sigma)$ -norm controls and found that approximations converged in a pointwise sense, even for $\lambda < 1$. Of course, for the same initial conditions, the minimum $H^1(\Sigma)$ -norm control is totally different from the minimum $L^2(\Sigma)$ -norm control, e.g., compare Figure 3 with Figures 6 and 7.

We now ask the question: can we recover a good pointwise approximation to the minimum $L^2(\Sigma)$ -norm control even for $\lambda < 1$? To this end, we next consider approximations of the particular solution of the exact controllability problem (1) that minimizes the norm-like functional $\mathcal{K}_\epsilon(\cdot)$ given in (12). Computational results are given in Table 5 and in Figures 8 and 9. Of course, with $\epsilon = 0$ and $\epsilon = 1$ we recover the results for the minimum $L^2(\Sigma)$ -norm control and minimum $H^1(\Sigma)$ -norm control cases, respectively.

From Table 5, we see that as $\epsilon \rightarrow 0$ the $L^2(\Sigma)$ -norm of the approximate control g^h converges to the value obtained with $\epsilon = 0$. This gives evidence of the convergence (in $L^2(\Sigma)$ and $L^2(Q)$) of the sequence of approximate controls and states as $\epsilon \rightarrow 0$.

TABLE 5. Norms of the approximate controls and states that minimize the functional (12) for the examples with initial data (16); $h = 1/128$ and $\lambda = 7/8$.

		$\epsilon = 0$	$\epsilon = 0.001$	$\epsilon = 0.01$	$\epsilon = 0.1$	$\epsilon = 1$
I	$\ g^h\ _{L^2(\Sigma)}$	0.6036	0.6100	0.6348	0.7172	0.7824
	$\ u^h\ _{L^2(Q)}$	0.7140	0.7185	0.7206	0.7774	0.8230
II	$\ g^h\ _{L^2(\Sigma)}$	1.0933	1.1914	1.3665	1.5153	2.2283
	$\ u^h\ _{L^2(Q)}$	1.1964	1.2242	1.2709	1.3311	1.7084

Figures 8 to 10 illustrate the pointwise behavior of the approximate solutions. The left-hand plots in Figures 8 to 9 show the behavior of the approximate controls obtained for the fixed value of $h = 1/128$ and for several values of ϵ . We see that for relatively large values of ϵ , the approximate controls are smooth. However, for the smallest value $\epsilon = 0.001$, we see evidence of oscillatory behavior but we also see that the approximate controls very closely resemble the piecewise smooth controls obtained with $\epsilon = 0$ and $\lambda = 1$, i.e., compare with Figures 3. The right-hand plots in Figures 8 to 9 show the behavior of the approximate controls obtained for a fixed value of $\epsilon = 0.001$ and for several values of h . We now see that for relatively large values of h , the approximate controls are oscillatory in nature but that as the grid size decreases, the amplitude of the oscillations decrease. These observations lead us to conclude that for any fixed value of $\epsilon > 0$, we should be able to obtain piecewise smooth (non-oscillatory) approximate controls simply by choosing a sufficiently small grid size h . Since we also conclude that we can approximate the minimum $L^2(\Sigma)$ -norm control as well as we like by choosing ϵ sufficiently small in the functional given in (12), we finally conclude that for any $\lambda < 1$ we can obtain pointwise convergent approximations to piecewise smooth minimum $L^2(\Sigma)$ -norm controls by first choosing ϵ sufficiently small and then choosing h small enough so that no oscillations occur. This last conclusion is illustrated in Figure 10 in which we compare the results obtained by minimizing the the functional given in (12) with $\lambda = 7/8$, $\epsilon = 0.001$, and $h = 1/256$ with the baseline solution g^* obtained by minimizing the $L^2(\Sigma)$ -norm with $\lambda = 1$ and $h = 1/256$. The pointwise convergence of the former is evident.

4. Concluding remarks

We have studied, both analytically and computationally, optimization-based numerical methods for exact boundary controllability problems for the wave equation in one space dimension. In general, our methodology produces convergent, in an L^2 sense, approximations without the need for regularization. Pointwise convergence is obtained for smooth enough controls and a simple regularization procedure also results in pointwise convergent approximations, even for controls that are only piecewise smooth. The methodology is very flexible in that one may treat essentially arbitrary controls and arbitrary minimization criteria.

The optimization-based numerical methods can easily be extended to the wave equation in higher dimensions. In this case, efficient implementation strategies play an even larger role since our methodology involves the solution of space-time coupled problems.

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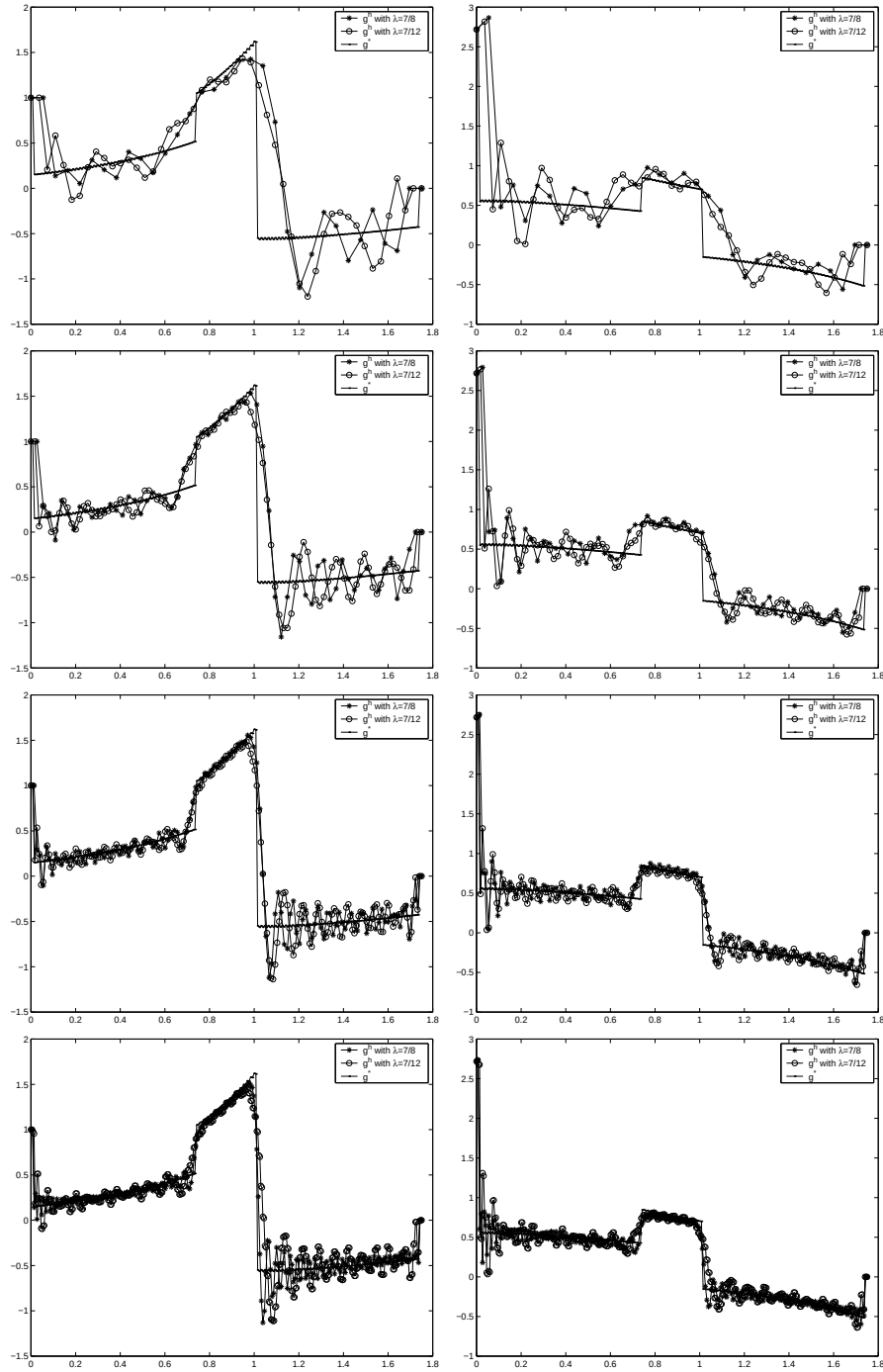


FIGURE 5. The approximate minimum $L^2(\Sigma)$ -norm Dirichlet control g^h vs. time for the example with initial data (16-II) at $x = 0$ (left) and $x = 1$ (right). From top to bottom: $h = 1/16$; $h = 1/32$; $h = 1/64$; $h = 1/128$.

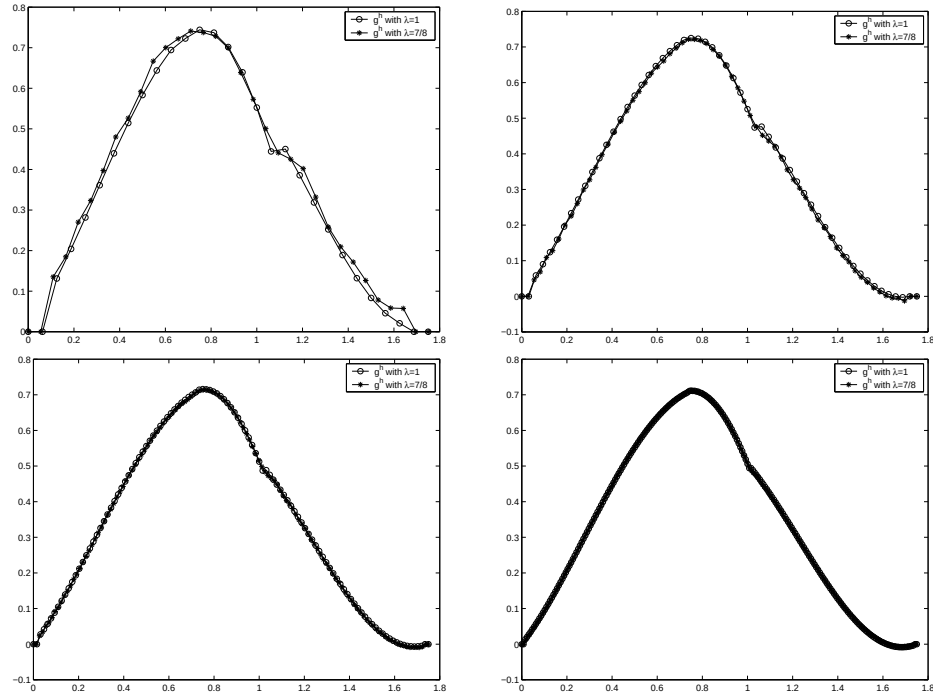


FIGURE 6. The approximate minimum $H^1(\Sigma)$ -norm Dirichlet controls g^h vs. time at $x = 0$ (and $x = 1$) for the initial data (16-I). Top-left: $h = 1/16$; top-right: $h = 1/32$; bottom-left: $h = 1/64$; bottom-right: $h = 1/128$.

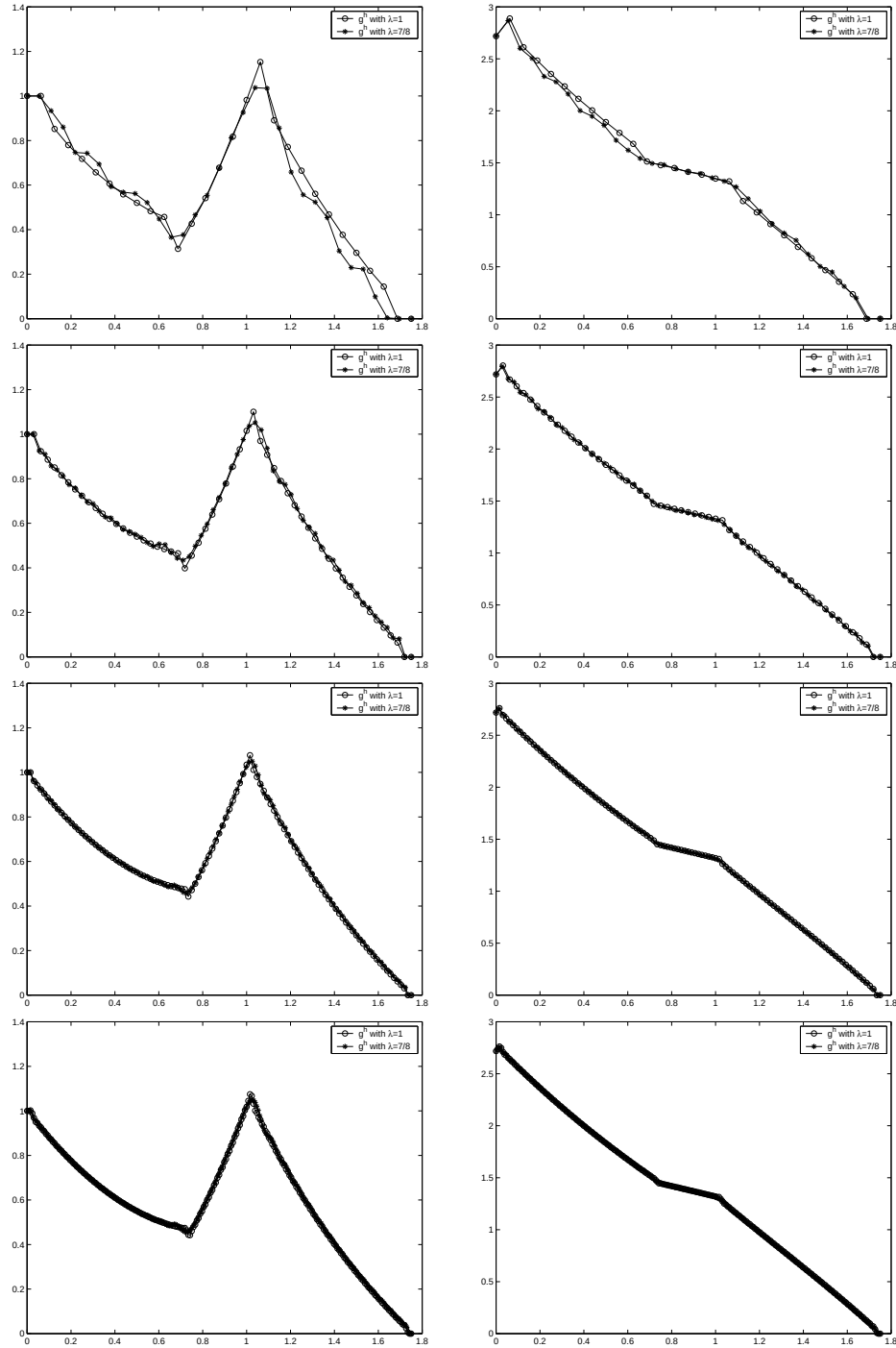


FIGURE 7. The approximate minimum $H^1(\Sigma)$ -norm Dirichlet control g^h vs. time for the example with initial data (16-II) at $x = 0$ (left) and $x = 1$ (right). From top to bottom: $h = 1/16$; $h = 1/32$; $h = 1/64$; $h = 1/128$.

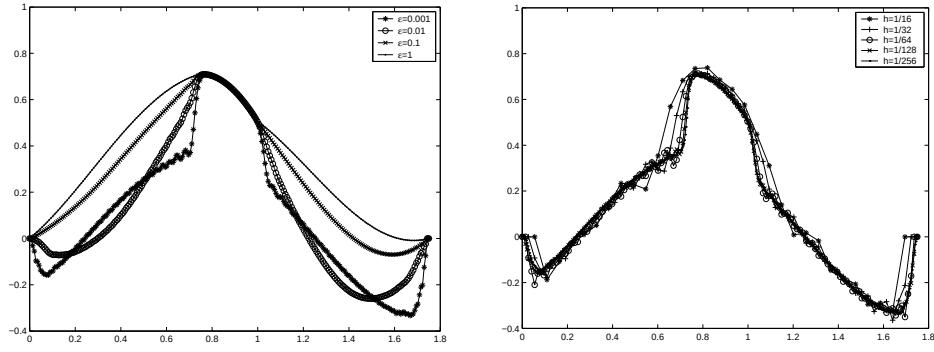


FIGURE 8. Approximate controls at $x = 0$ (and $x = 1$) that minimize the functional (12) vs. time for the examples with initial data (16-I) for $\lambda = 7/8$. Left: $h = 1/128$; right: $\epsilon = 0.001$.

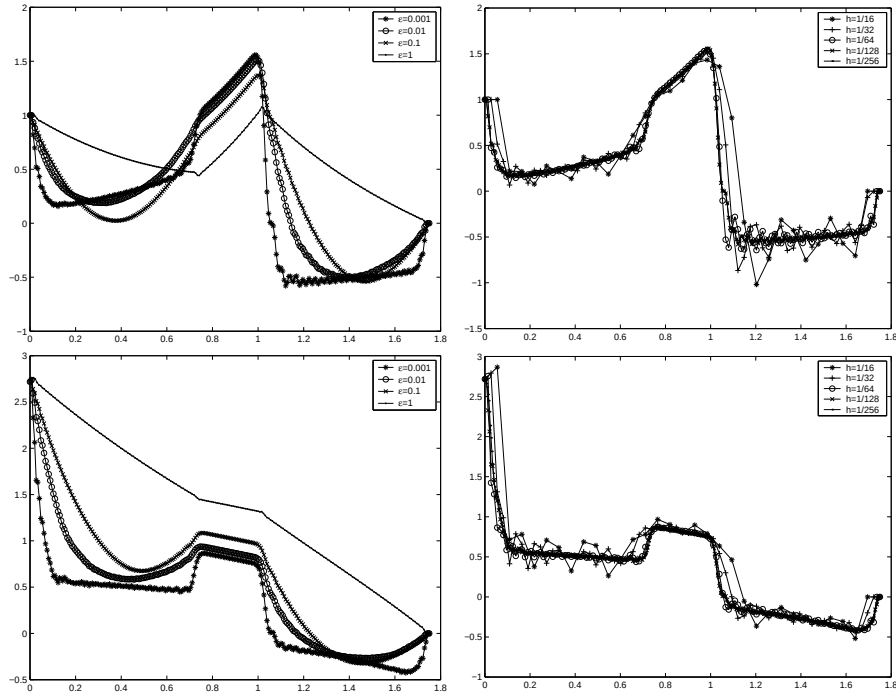


FIGURE 9. Approximate controls at $x = 0$ (top) and $x = 1$ (bottom) that minimize the functional (12) vs. time for the examples with initial data (16-II) for $\lambda = 7/8$. Left: $h = 1/128$; right: $\epsilon = 0.001$.

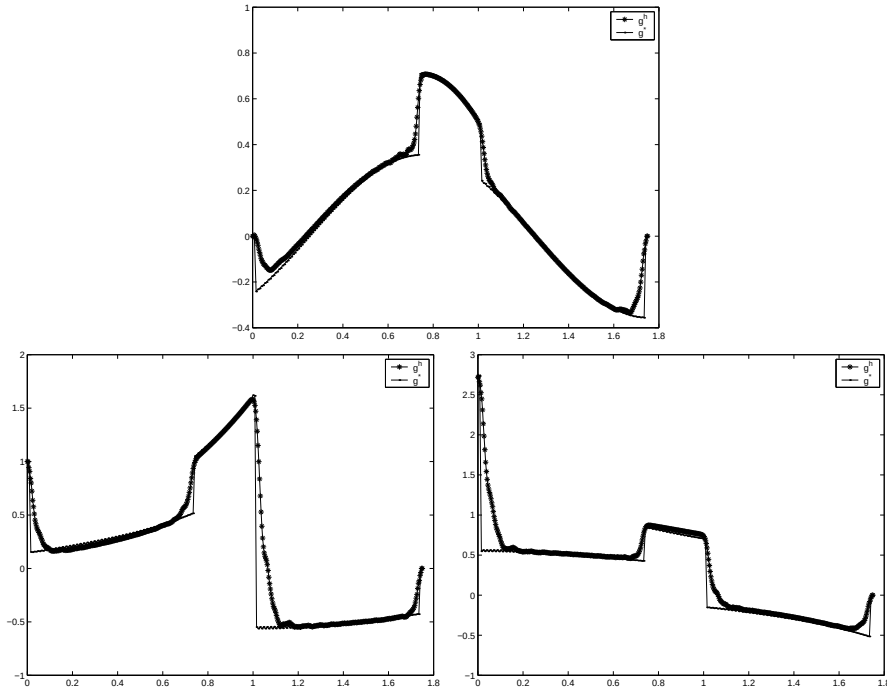


FIGURE 10. Approximate controls vs. time for the examples with initial data (16); g^h is for the minimization of the functional (12) with $\epsilon = 0.001$, $\lambda = 7/8$, and $h = 1/256$; g^* is the baseline approximate control that minimizes the $L^2(\Sigma)$ -norm with $\lambda = 1$ and $h = 1/256$. Top: approximate controls at $x = 0$ (and $x = 1$) for the initial data (16-I); bottom-left and bottom-right: approximate controls at $x = 0$ and $x = 1$, respectively, for the initial data (16-II).